

EPA Air Quality Datasets

Summary

This dataset was prepared for Environmental Data Analytics (ENV 700) at Duke University, Fall 2026

The dataset contains data from air quality monitoring of PM2.5 and ozone in North Carolina in 2024 and 2025.

Database Information

Data were collected using the Download Daily Data tool (<https://www.epa.gov/outdoor-air-quality-data/download-daily-data>).

The following selections were made:

- PM2.5 and Ozone (Pollutant)
- 2024 and 2025 (Year)
- North Carolina (Geographic Area)
- Download CSV (spreadsheet)

csv files were saved as `EPAair_03_NC2024_raw.csv`, `EPAair_03_NC2025_raw.csv`, `EPAair_PM25_NC2024_raw.csv`, and `EPAair_PM25_NC2025_raw.csv`.

Data were accessed 2026-06-09.

Data Content Information

Information gathered from: <https://www.epa.gov/outdoor-air-quality-data/air-data-basic-information> and https://aqs.epa.gov/aqsweb/documents/AQS_Format.html

Column names without descriptors are self-explanatory.

Date: month/day/year

Source: AQS (Air Quality System) or AirNow

Site ID: A unique number within the county identifying the site.

POC: "Parameter Occurrence Code" used to distinguish different instruments that measure the same parameter at the same site.

Daily Mean PM2.5 Concentration: numeric value

Daily Max 8-hour Ozone Concentration: numeric value

Units: units for concentration

Daily_AQI_VALUE: Air quality index (range 0-500). Levels:

0-50: Good (green)

51-100: Moderate (yellow)

101-150: Unhealthy for sensitive groups (orange)

151-200: Unhealthy (red)

201-300: Very unhealthy (purple)

301-500: Hazardous (maroon)

Site Name
DAILY_OBS_COUNT: number of observations per day
PERCENT_COMPLETE
AQS_PARAMETER_CODE
AQS_PARAMETER_DESC
CBSA_CODE
CBSA_NAME
STATE_CODE
STATE
COUNTY_CODE
COUNTY
SITE_LATITUDE
SITE_LONGITUDE

Additional Information and Support

For more information, please contact the data assembler, **John Fay** (john.fay@duke.edu)